Business Requirements Document: Online Student Leave Management System

1. Problem Statement

Students apply for leave on paper forms submitted to class coordinators. After submission, the request cannot be tracked, decisions are often communicated late, and staff must search physical files whenever a past leave record is needed.

- **No status visibility:** students do not know whether leave is approved before they miss classes or exams.
- **Slow approvals:** a decision depends on the coordinator being physically available.
- **Lost time and records:** staff search old files manually; forms can be lost or damaged.
- **No data:** management cannot see leave patterns by student, class or department.

2. Business Objectives

Objective	Success measure	Target
Digitise all leave requests	% of requests submitted online	100% within one semester; paper forms withdrawn
Give students real-time status	Requests with live status and notifications	100%
Reduce approval time	Average time from submission to decision	⩽ 24 working hours for standard leave
Remove manual record searching	Time to find any past leave record	Under 1 minute
Provide leave reports to management	Standard reports available	Monthly reports by student, class and department

3. Project Scope

3.1 In scope

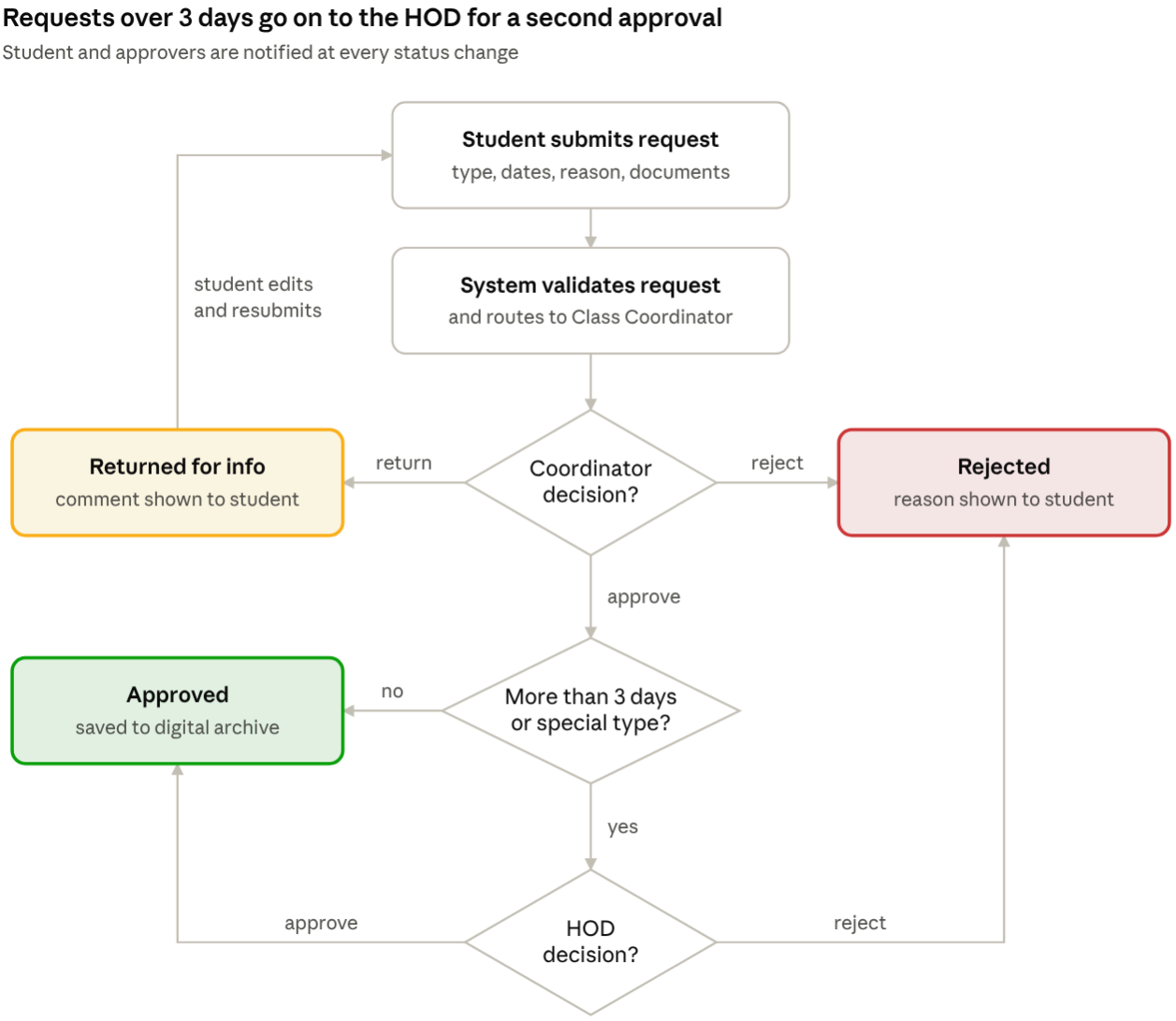
- Login with college credentials and role-based access (student, coordinator, HOD, office staff, admin)
- Online leave application with leave type, dates, reason and document upload

- Two-level approval: Class Coordinator, then HOD for leave over 3 days or special types
- Real-time status tracking, email and in-app notifications, reminders and escalation
- Searchable digital archive and standard leave reports
- Responsive web access on computers and mobile browsers

3.2 Out of scope

- Staff and faculty leave
- Automatic attendance marking or write-back to the attendance module
- Native Android/iOS apps and SMS/WhatsApp alerts
- Hostel leave and gate passes
- Digitising paper records older than the current academic year

3.3 To-be process flow



To-be leave approval flow · 3 decisions, 3 outcomes

A request idle with an approver for 24 working hours triggers a reminder and is escalated to the HOD after 48 hours (FR-07); the student can cancel any request before its start date (FR-06).

4. Stakeholder Register

Stakeholder	Responsibilities	Interest / Influence	Engagement
Principal / Dean (Sponsor)	Approves budget, leave policy and this BRD; resolves escalations	High / High	Steering updates every two weeks; formal sign-off
Head of Department (HOD)	Sets department approval rules; approves long or special leave; reviews reports	High / High	Interviews, workflow review, UAT
Class Coordinator	First-level approver; verifies documents; communicates decisions	High / Medium	Workshops, prototype review, UAT
Students	Submit accurate requests with documents; track status; give feedback	High / Medium	Survey, focus group, pilot class
Parents / Guardians	Receive leave notifications	Medium / Low	College circular and email
Administrative Office Staff	Maintain records; prepare attendance and audit reports	High / Medium	Process interviews, report review, UAT
IT Department	Hosting, login and ERP integration, security, backups, support	Medium / High	Technical workshops, security review

5. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall allow users to log in with college credentials and show features based on their role.	Must

ID	Requirement	Priority
FR-02	The system shall allow a student to submit a leave request with leave type, dates (full/half day), reason and supporting documents (PDF/JPG/PNG, max 5 MB).	Must
FR-03	The system shall validate requests: no end date before start date, no overlapping requests, and a medical certificate for medical leave over 2 days.	Must
FR-04	The system shall route each request to the student's class coordinator, and to the HOD when leave exceeds 3 days or is of a special type.	Must
FR-05	The system shall allow approvers to approve, reject (reason required) or return a request for more information.	Must
FR-06	The system shall show students the real-time status and full history of each request, and allow cancellation before the leave starts.	Must
FR-07	The system shall send email and in-app notifications on every status change, remind approvers after 24 working hours and escalate to the HOD after 48 hours.	Must
FR-08	The system shall allow authorised staff to search leave records and export reports by student, class, department, type and date range.	Should

6. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Pages and searches load within 3 seconds for 95% of requests with 500 concurrent users.
NFR-02	Security	Role-based access, HTTPS encryption, 15-minute idle timeout; medical documents visible only to the student and approvers; compliant with the DPDP Act, 2023.
NFR-03	Availability	99.5% uptime during college hours, with daily backups.

ID	Category	Requirement
NFR-04	Usability	A first-time student can submit a request in under 3 minutes on a mobile browser without training.

7. Assumptions and Constraints

Assumptions

- Every student and staff member has an active college email ID and login.
- Students have access to a smartphone or computer with internet.
- Student-to-class and class-to-coordinator mapping in the college ERP is accurate each semester.
- The leave policy (types, limits, document rules) will be approved by the sponsor before design.
- Management will stop accepting paper forms after go-live.

Constraints

- **Budget:** fixed college IT budget; open-source or existing licensed tools preferred.
- **Time:** go-live before the next semester, after a one-department pilot.
- **Technology:** must integrate with the existing ERP and college login.
- **Regulatory:** must follow the DPDP Act, 2023 and university record-keeping norms.
- **Policy:** the system follows the existing leave and attendance policy; it does not change it.

8. Stakeholder Communication and Requirement-Gathering Plan

Activity / technique	Stakeholders	Purpose	Timing
Document analysis	Admin Office	Study the paper form, leave policy and past records	Week 1
Interviews	Principal, HODs, IT Head	Goals, approval rules, constraints, integration needs	Week 1
Observation	Class coordinators, office staff	See the current process; measure approval and search time	Week 1–2

Activity / technique	Stakeholders	Purpose	Timing
Online survey	All students	Pain points, device access, notification preferences	Week 2
Requirements workshop	Coordinators, HODs, office staff, IT	Agree workflow, statuses, validation rules and reports	Week 3
Prototype walkthrough	Students, coordinators, HODs	Validate screens and flow before development	Week 4
Progress update	Sponsor, HODs	Status, risks, decisions needed	Every two weeks
BRD sign-off	Sponsor, HODs, IT Head	Approve scope and requirements	End of Week 4

9. User Stories and Acceptance Criteria

US-01 – Submit a leave request

As a student, I want to submit a leave request online with supporting documents, so that I do not have to find my coordinator in person or fill a paper form.

- Given I am logged in, when I open "Apply for Leave", then I see fields for leave type, dates, reason and document upload.
- Given all mandatory fields are valid, when I submit, then the request gets a unique ID, status "Submitted", and is sent to my class coordinator.
- Given the end date is before the start date or overlaps another request, when I submit, then the system blocks it and shows an error.
- Given my request is submitted, then I and my coordinator receive a notification within 1 minute.

US-02 – Approve or reject a request

As a class coordinator, I want to see all pending requests for my class and approve, reject or return them with comments, so that I can decide quickly without paper forms.

- Given I am logged in, when I open my dashboard, then I see only my class's pending requests, oldest first.
- Given I approve a request over 3 days, then its status becomes "Pending HOD Approval" and it moves to the HOD.
- Given I reject or return a request, when the comment is empty, then the system does not allow the action.

- Given a request is pending with me for 24 working hours, then I receive a reminder.

US-03 – Track request status

As a student, I want to track the status of my leave requests, so that I know whether my leave is approved before I miss classes or exams.

- Given I open "My Leave Requests", then I see each request's ID, type, dates, current status and current approver.
- Given I open a request, then I see a timeline of every action with date, time, approver and comments.
- Given the status changes, then I receive an email and in-app notification.
- Given my leave has not started, when I cancel it, then the status becomes "Cancelled" and the approver is notified.